

Human-Computer Interaction

Accessibility, Aging & AI

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Today's Agenda

- Topic overview: *Accessibility, Aging & AI*
- The frame: *ability-based design* → *adaptive AI mediation*
- Provocation: *AI-as-liberation or AI-as-erasure?*
- Discussion

This term's frame

Enduring fundamental → How AI is changing it → Depth (a claim to argue with)

- **Fundamental:** design for the full range of human ability; access is *relational*
- **How AI changes it:** personalized **mediation** becomes possible, but the same systems can **encode ableism**
- **Depth: Deep:** accommodation shifts from fixed fixes to *adaptive mediation*

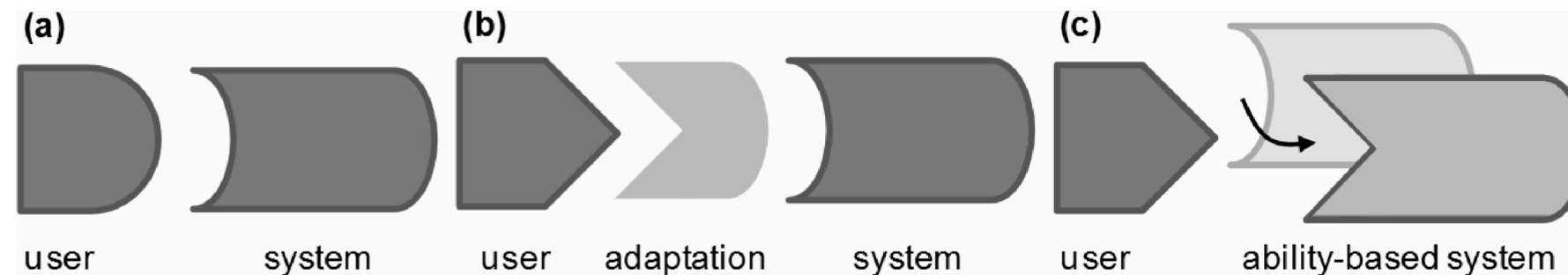
This week's readings

- **Required**
 - Wobbrock et al., *Ability-Based Design* (TACCESS 2011)
 - Glazko et al., *An Autoethnographic Case Study of GenAI's Utility for Accessibility* (ASSETS 2023)
- **Optional**: Morris, *AI and Accessibility* (CACM 2020) · Whittaker, Morris et al., *Disability, Bias, and AI* (AI Now, 2019)

Topic overview: ***Accessibility, Aging &*** ***AI***

Ability-based design¹

- **Ability:** "focus on ability not dis-ability, striving to leverage all that users can do"
- **Accountability:** "respond to poor performance by changing systems, not users"



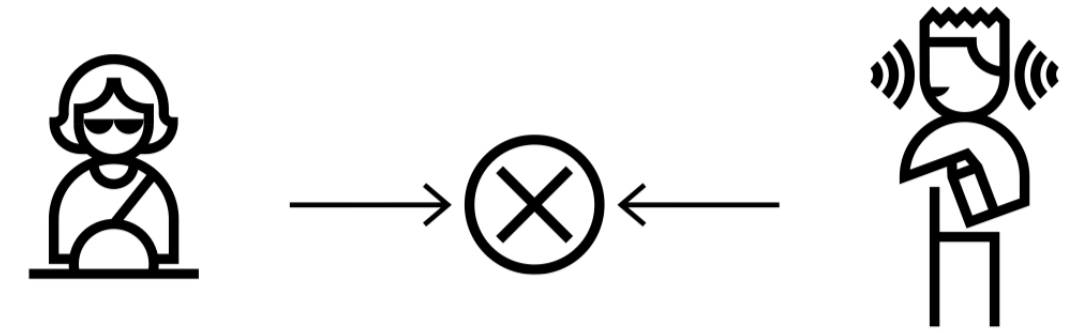
¹Wobbrock, Kane, Gajos, Harada & Froehlich (2011). Ability-Based Design: Concept, Principles and Examples. *ACM TACCESS*, 3(3), Fig. 2 and Table 1.

Access is relational²

Disability "results from the interaction between health conditions... and a range of environmental and personal factors."

Change the context and the barrier can disappear: the design carries the accommodation.

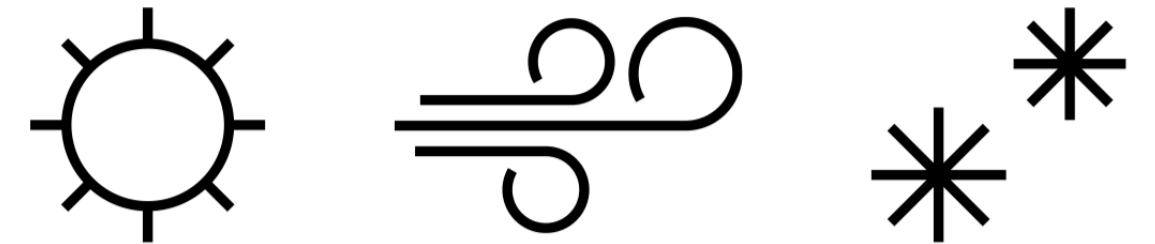
Between
humans



Can't type

Can't hear

Human+
enviroment

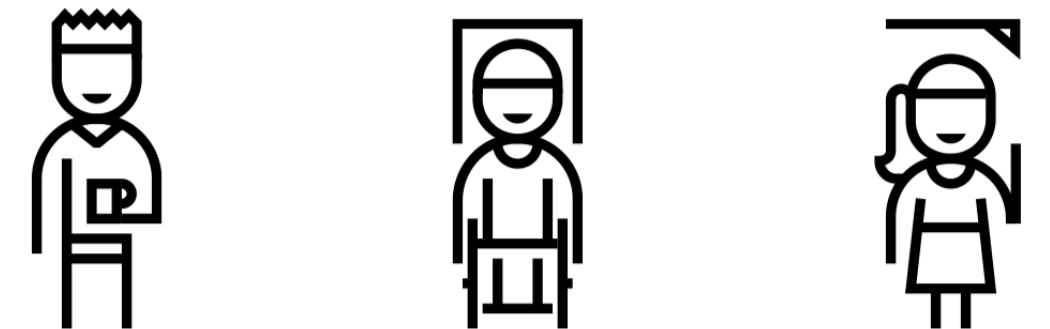


Glare from sun

Windy

Cold

Human+
object



Left-handed user

Narrow door

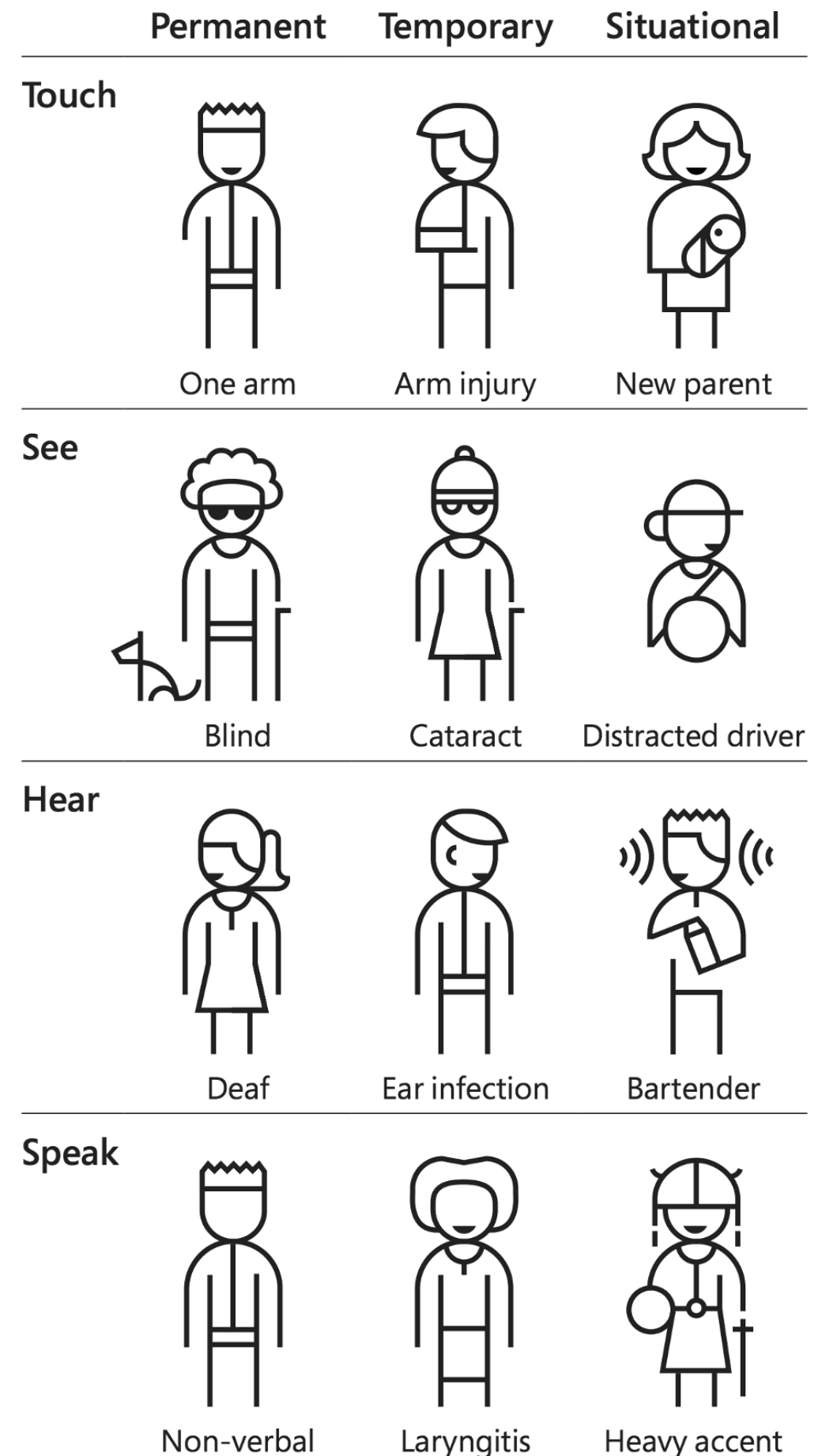
Tall shelf

²WHO (2023). *Disability* fact sheet. Figure: Microsoft Inclusive Design Toolkit.

Ability changes over time³

- **1.3 billion** people, "1 in 6 of us," experience significant disability
- The number is growing, in part because of **people living longer**
- **Temporary** and **situational** impairments touch everyone

³WHO (2023). *Disability* fact sheet. Figure: Microsoft Inclusive Design Toolkit (persona spectrum).



What AI changes: personalized mediation

On-demand, bespoke adaptation is finally possible.

- Multimodal **description** and scene understanding
- Real-time **captioning** and translation
- **Simplification** and reading support
- **Agentic** mediation that acts on the user's behalf

👉 Accommodation moves from *fixed add-ons* to *adaptive mediation*.

The duality, first-person⁴

A three-month autoethnography by researchers **with and without** disabilities.

- **Useful:** references, social messages, drafting, code for a blind developer
- **But** it "almost always required human verification"
- It would "repeatedly provide a false solution" while "stating confidently that it understood accessibility guidelines"

⁴Glazko, Yamagami, Desai, Mack, Potluri, Xu & Mankoff (2023). An Autoethnographic Case Study of Generative AI's Utility for Accessibility. *ASSETS '23*.

AI can encode ableism⁵ ⁶

The same models that mediate also **allocate** and **represent**.

- GPT-4 ranked résumés **lower** when they listed disability-related honors⁵
- Systems trained on ableist data reproduce it at scale⁶
- "Better tools" and "new harm" are **not** separate stories

⁵ Glazko, Mohammed, Kosa, Potluri & Mankoff (2024). Identifying and Improving Disability Bias in GPT-Based Resume Screening. *FAccT '24*.

⁶ Whittaker, Morris et al. (2019). *Disability, Bias, and AI*. AI Now Institute. (optional)

The failure-cost asymmetry

Not all errors cost the same.

*A wrong scene description, **trusted** by a blind user, is not a small error: it is a decision made on false information.*

- Mean accuracy hides the **distribution** and **consequence** of failure
- Evaluate by *who bears the cost of being wrong*

Fold-in: participatory & justice design⁷

Not just an attribute of the studies we cite, but a **method**.

- Scandinavian **participatory design**: those affected are co-designers
- **Co-design** and **design justice**: shift who holds design power
- The operational form of "*nothing about us without us*"

⁷ Costanza-Chock, S. (2020). *Design Justice: Community-Led Practices to Build the Worlds We Need*. MIT Press. Open access: designjustice.mitpress.mit.edu

Guardrails coda

- **Whose descriptions count**: who authored the mediation, and for whom?
- **Verifiability**: can the user check what the system asserts?
- **Autonomy vs. dependence**: does the tool extend agency or replace it?

Liberation or *erasure*?

The tension, today

- **Liberation**: bespoke access at scale
 - **Erasure**: ableism at scale, and user agency displaced
 - **Wobbrock**: change the system, not the user
 - **Glazko**: useful, but only if you can verify it
- 👉 The test: does the system *center* disabled users, or *speak over* them?

Your provocation (*post one slide*)

Pick one:

1. **Provocation**: when does personalized AI mediation *empower* disabled users, and when does it *displace* their agency?
2. **Critical artifact**: an AI accessibility tool or output that helps *and* misleads (an auto-caption, a scene description)
3. **Design / policy**: one principle for "*nothing about us without us*" in AI-assistive design

Discussion

Today's roles

Reporter

Skeptic

Connector

Seat 5

Seat 6

Seat 1

Same in every pod. **Seats 2, 3 and 4** have no role today; you are still in the argument.

Discussion Format

In your pod, on your pod monitor: ~30 minutes.

- **Everyone shows their slide:** about 2 minutes each, cast from your own laptop
- **Skeptic** puts a counterargument on the table · **Connector** ties it to an earlier week
- **Reporter** writes the pod's three lines (agreed, split, unresolved) and delivers them to the room

What's Next?

- **Friday: Method-Design Crit (M6)**
 - **Due 9:30 am:** Step-by-Step Experimental Design (*individual*); bring it, it's what you crit
 - **If you can:** LFH **Ch. 13** (Measuring the human) · optional Ch. 5 (Surveys)